

A Finite-Dimensional Operator-Algebra Case of the Clouatre–Ramsey Envelope Conjecture

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Source Problem

Clouatre and Ramsey define, for an operator space $\mathcal{M}$, a completely isometric representation $\mu : \mathcal{M} \rightarrow B(\mathcal{H}_\mu)$, and $r \geq 1$, a C_r^* -envelope $C_{e,r}^*(\mathcal{M}, \mu)$. After proving in Theorem 8.9 that there is always a surjective $*$ -homomorphism

$$\Gamma_r : C_{e,1}^*(\mathcal{M}, \mu) \rightarrow C_{e,r}^*(\mathcal{M}, \mu), \quad \Gamma_r \circ \varepsilon_{\mu,1} = \varepsilon_{\mu,r},$$

they conjecture that $C_{e,1}^*(\mathcal{M}, \mu)$ and $C_{e,r}^*(\mathcal{M}, \mu)$ are $*$ -isomorphic for every $r \geq 1$. The conjecture appears on PDF page 43 of arXiv:1703.02924 and in the parsed source at `CBchoquet_arXiv_v3_2018Feb28.tex:24`

Conjecture. Let $\mathcal{M}$ be an operator space, let $\mu : \mathcal{M} \rightarrow B(\mathcal{H}_\mu)$ be a completely isometric linear map and let $r \geq 1$. Then, the C^* -algebras $C_{e,1}^*(\mathcal{M}, \mu)$ and $C_{e,r}^*(\mathcal{M}, \mu)$ are $*$ -isomorphic.

We close the paper by verifying this conjecture in some cases of interest. We already observed that if $C^*(\mu(\mathcal{M}))$ is simple, then $C_{e,r}^*(\mathcal{M}, \mu)$ and $C^*(\mu(\mathcal{M}))$ are $*$ -isomorphic for every $r \geq 1$. On the other hand, if we assume that $C^*(\mu(\mathcal{M}))$ contains the ideal $\mathcal{K}$ of compact operators on $\mathcal{H}_\mu$, then we can also achieve some partial success. The following can be viewed as a variation of an important fact called *Arveson's boundary theorem* [5].

Theorem 8.11. *Let $\mathcal{M}$ be an operator space and let $\mu : \mathcal{M} \rightarrow B(\mathcal{H}_\mu)$ be a completely isometric linear map such that $C^*(\mu(\mathcal{M}))$ contains the ideal $\mathcal{K}$ of compact operators on $\mathcal{H}_\mu$. Let $q : C^*(\mu(\mathcal{M})) \rightarrow C^*(\mu(\mathcal{M}))/\mathcal{K}$ denote the quotient map. Let $r \geq 1$ and assume that there is an integer $n \in \mathbb{N}$ and an element $a \in M_n(\mu(\mathcal{M}))$ such that $\|q^{(n)}(a)\| < r^{-1}\|a\|$. Then, the map $\varepsilon_{\mu,r} : C^*(\mu(\mathcal{M})) \rightarrow C_{e,r}^*(\mathcal{M}, \mu)$ is a $*$ -isomorphism.*

Proof. Basic representation theory for C^* -algebras [21] stipulates that the $*$ -homomorphism

$$\varepsilon_{\mu,r} : C^*(\mu(\mathcal{M})) \rightarrow B(\mathfrak{H}^{r,\mu})$$

is unitarily equivalent to

$$\text{id}^{(\mathfrak{c})} \oplus \sigma \circ q$$

for some cardinal number $\mathfrak{c}$ and some $*$ -homomorphism σ on $C^*(\mu(\mathcal{M}))/\mathcal{K}$. By assumption, we see that $\|(\sigma \circ q)^{(n)}(a)\| < r^{-1}\|a\|$. In view of Corollary 8.1, we conclude that $\mathfrak{c} \neq 0$. In particular $\varepsilon_{\mu,r}$ is injective and thus

$$\varepsilon_{\mu,r} : C^*(\mu(\mathcal{M})) \rightarrow C_{e,r}^*(\mathcal{M}, \mu)$$

is a $*$ -isomorphism □

Result

The source proves the conjecture when $C^*(\mu(\mathcal{M}))$ is simple, for a compact-operator boundary-theorem situation, and for uniform algebras. This packet adds a finite-dimensional operator-algebra case.

Theorem 1. *Let $\mathcal{A}$ be a unital operator algebra and let*

$$\alpha : \mathcal{A} \rightarrow B(\mathcal{H}_\alpha)$$

be a unital completely isometric homomorphism. Assume that

$$\mathfrak{B} = C_{e,1}^*(\mathcal{A}, \alpha)$$

is finite-dimensional. Then for every $r \geq 1$, the Clouatre–Ramsey comparison map

$$\Gamma_r : \mathfrak{B} \rightarrow C_{e,r}^*(\mathcal{A}, \alpha)$$

is injective. Consequently $C_{e,1}^(\mathcal{A}, \alpha)$ and $C_{e,r}^*(\mathcal{A}, \alpha)$ are $*$ -isomorphic for every $r \geq 1$.*

Proof Intuition

Because $\mathfrak{B}$ is finite-dimensional, a nonzero kernel of Γ_r must kill at least one matrix-algebra summand. But the summands of the ordinary $C_{e,1}^*$ -envelope come from boundary representations. In finite dimension, boundary representations are peaking: there is a matrix over the operator algebra whose norm is attained on that summand and is strictly smaller on all other summands. The operator-algebra assumption is exactly what lets us tensor this peaking matrix with itself while staying inside matrices over the same algebra. Tensor powers drive the off-summand norm ratio below $1/r$, whereas Clouatre–Ramsey’s universal lower bound for $\varepsilon_{\alpha,r}$ says no matrix over $\alpha(\mathcal{A})$ can lose more than the factor $1/r$. The two estimates are incompatible.

Peaking Input

We use the following standard finite-dimensional consequence of non-commutative peaking theory. Let E be a unital operator algebra whose C^* -envelope $\mathfrak{B} = C_e^*(E)$ is finite-dimensional. Write

$$\mathfrak{B} = \bigoplus_{j=1}^m \mathfrak{B}_j$$

as a direct sum of full matrix algebras, and let $\pi_j : \mathfrak{B} \rightarrow \mathfrak{B}_j$ be the coordinate irreducible representations. Then each π_j is a local peak representation for E . Equivalently, for each j there are $n \geq 1$ and $X \in M_n(E)$ such that

$$\|\pi_j^{(n)}(X)\| = 1, \quad c_j := \max_{i \neq j} \|\pi_i^{(n)}(X)\| < 1.$$

For the present setting, $E = \varepsilon_{\alpha,1}(\alpha(\mathcal{A}))$. The reason the hypothesis applies is that the source construction of $C_{e,1}^*$ is built from $\mathcal{P}_1$ -extremal, equivalently completely positive boundary, representations; reducing summands of such extremals remain extremal [1, Lemma 7.2]. Since $\mathfrak{B}$ is finite-dimensional, every irreducible coordinate is finite-dimensional and isolated. Clouatre–Thompson’s peaking results for finite-dimensional boundary representations then give the displayed local peak matrix [2, Sections 3–4]; see also their IMRN abstract, which describes the isolated-spectrum mechanism for finite-dimensional boundary representations.

Proof

Put

$$E = \varepsilon_{\alpha,1}(\alpha(\mathcal{A})) \subset \mathfrak{B}.$$

Since $\mathcal{A}$ is an operator algebra and $\varepsilon_{\alpha,1}$ is the restriction of a $*$ -homomorphism on $C^*(\alpha(\mathcal{A}))$, E is a unital operator algebra generating $\mathfrak{B}$. By Corollary 8.1 of [1] with $r = 1$, $\varepsilon_{\alpha,1}$ is completely isometric on $\alpha(\mathcal{A})$.

Fix $r \geq 1$, and suppose toward a contradiction that

$$J = \ker \Gamma_r$$

is nonzero. Since $\mathfrak{B}$ is finite-dimensional, after writing $\mathfrak{B} = \bigoplus_{j=1}^m \mathfrak{B}_j$, the ideal J contains at least one summand. Choose such a summand and call it $\mathfrak{B}_j$.

By the peaking input, there are $n \geq 1$ and $X = [x_{ab}] \in M_n(E)$ such that

$$\|\pi_j^{(n)}(X)\| = 1, \quad c := \max_{i \neq j} \|\pi_i^{(n)}(X)\| < 1.$$

Choose $k \geq 1$ so large that $c^k < r^{-1}$. Because E is an algebra, the algebraic matrix tensor power

$$X^{\otimes_E k} \in M_{n^k}(E)$$

is again a matrix over E : its $(\mathbf{a}, \mathbf{b})$ -entry is

$$x_{a_1 b_1} x_{a_2 b_2} \cdots x_{a_k b_k}, \quad \mathbf{a} = (a_1, \dots, a_k), \quad \mathbf{b} = (b_1, \dots, b_k).$$

For every coordinate representation π_i ,

$$\pi_i^{(n^k)}(X^{\otimes_E k}) = \pi_i^{(n)}(X) \otimes \cdots \otimes \pi_i^{(n)}(X),$$

so

$$\|\pi_j^{(n^k)}(X^{\otimes_E k})\| = 1, \quad \max_{i \neq j} \|\pi_i^{(n^k)}(X^{\otimes_E k})\| \leq c^k < r^{-1}.$$

Thus $\|X^{\otimes_E k}\| = 1$. Since Γ_r kills the j -summand,

$$\|\Gamma_r^{(n^k)}(X^{\otimes_E k})\| \leq c^k < r^{-1}.$$

Now choose $A_k \in M_{n^k}(\alpha(\mathcal{A}))$ such that

$$\varepsilon_{\alpha,1}^{(n^k)}(A_k) = X^{\otimes_E k}.$$

The complete isometry of $\varepsilon_{\alpha,1}$ gives $\|A_k\| = 1$. On the other hand, $\Gamma_r \circ \varepsilon_{\alpha,1} = \varepsilon_{\alpha,r}$, hence

$$\|\varepsilon_{\alpha,r}^{(n^k)}(A_k)\| = \|\Gamma_r^{(n^k)}(X^{\otimes_E k})\| < r^{-1}.$$

This contradicts Corollary 8.1 of [1], which gives

$$\|\varepsilon_{\alpha,r}^{(n^k)}(A_k)\| \geq r^{-1} \|A_k\| = r^{-1}.$$

Therefore $\ker \Gamma_r = 0$. Since Γ_r is already surjective by Theorem 8.9 of [1], it is a $*$ -isomorphism.

Verification Notes

The proof does not claim that $\varepsilon_{\alpha,r}$ is a $\mathcal{P}$ -isomorphism on $\alpha(\mathcal{A})$, which is the sufficient condition isolated in Theorem 8.10 of [1]. Instead it proves injectivity of the already-existing quotient Γ_r directly from finite-dimensional peaking.

The tensor-power step is the only place where the operator-algebra hypothesis is used. For a general operator space, entries of $X^{\otimes_E k}$ would be products of entries of X , so there is no reason for the amplified matrix to remain in a matrix level over the original operator space.

Limitations and Novelty Check

This packet proves a subcase of the source conjecture, not the full operator-space conjecture. The finite-dimensional assumption is on the ordinary envelope $C_{e,1}^*(\mathcal{A}, \alpha)$; it is stronger than residual finite dimensionality and is not meant to cover the compact-operator or uniform-algebra families already handled in [1].

Cheap run indexes were searched for arXiv:1703.02924, the title, $C_{e,r}^*$, Clouatre–Ramsey, Choquet boundary, and finite-dimensional operator-algebra phrases; no duplicate packet for this result was found. A web literature search found the later Clouatre–Thompson peaking paper [2], which is used here as supporting theory rather than as an answer to the Clouatre–Ramsey $C_{e,r}^*$ -scale conjecture. The source paper itself lists the simple, compact-operator, and uniform-algebra cases, but not this finite-dimensional operator-algebra envelope case.

Human Review Recommendation

Review as a likely valid partial result. The priority checks are:

1. confirm the quoted finite-dimensional local-peaking input from Clouatre–Thompson applies to every irreducible summand of $C_{e,1}^*(\mathcal{A}, \alpha)$;
2. check the algebraic tensor-power matrix $X^{\otimes_E k}$ remains in $M_{n^k}(E)$ and has coordinate norms exactly amplified by powers;
3. verify that the lower bound in Corollary 8.1 of Clouatre–Ramsey applies at the matrix level used in the contradiction.

References

- [1] R. Clouatre and C. Ramsey, *A completely bounded non-commutative Choquet boundary for operator spaces*, International Mathematics Research Notices **2019**, no. 22, 6819–6886; arXiv:1703.02924.
- [2] R. Clouatre and I. Thompson, *Finite dimensionality in the non-commutative Choquet boundary: peaking phenomena and C^* -liminality*, International Mathematics Research Notices **2022**, no. 20, 16046–16093, DOI: [10.1093/imrn/rnab184](https://doi.org/10.1093/imrn/rnab184).
- [3] M. A. Dritschel and S. A. McCullough, *Boundary representations for families of representations of operator algebras and spaces*, Journal of Operator Theory **53** (2005), 159–167.